

ESpect Daily Startup Guide

Use this guide after the one-time VM setup has already completed successfully.

1. Start the VMs

Power on VM1 and VM2.

Expected addresses:

VM1: 192.168.160.128
VM2: 192.168.160.129

2. Confirm the persistent VM services

Run on VM1 and VM2:

```
sudo systemctl is-active strongswan  
sudo systemctl is-active espect-agent
```

Both commands must print:

active

If **espect-agent** is inactive:

```
sudo systemctl restart espect-agent  
sudo systemctl is-active espect-agent
```

If the service is missing, stop and use the first-time setup guide. Do not run pip or manually start **run_agent.py** during normal operation.

3. Confirm VM connectivity

Run on VM1:

```
ping -c 1 192.168.160.129  
nc -zv -w 2 192.168.160.128 9000  
nc -zv -w 2 192.168.160.129 9000
```

All checks must succeed. The inverse DNS warning from **nc** is harmless when the port reports **open**.

4. Start the Windows backend

Open PowerShell:

```
cd C:\Users\Subhadeep\ESpect\ipsec_analyzer\backend  
$connection = Get-NetTCPConnection -LocalPort 8000 -State Listen -ErrorAction SilentlyContinue  
if ($connection) { Stop-Process -Id $connection.OwningProcess -Force }  
.\venv\Scripts\python.exe -m uvicorn app.main:app --host 127.0.0.1 --port 8000
```

Keep this terminal open. In a second PowerShell window verify:

```
curl.exe -sS http://127.0.0.1:8000/health
```

Expected:

```
{"status": "ok"}
```

5. Start the Windows frontend

Open another PowerShell window:

```
cd C:\Users\Subhadeep\ESpect\ipsec_analyzer\frontend  
npm run dev
```

Keep the terminal open and open:

```
http://localhost:5173
```

6. Start a real capture

In the browser:

1. Open **Live capture**.
2. Click the main **Start capture** button once.
3. Wait approximately 30 seconds.
4. Wait for the dashboard to show completion and open the analysis results.

The dashboard starts the real VM1 workflow, which configures both agents, establishes IPsec, starts tcpdump, generates VoIP traffic, validates the PCAP, transfers it, and analyzes it.

Do not click individual workflow rows. They are status indicators only.

7. Expected result

A successful run contains:

```
capture_status: completed
```

The analysis should show:

- IKEv2 detected
- ESP detected
- Packets in both directions
- A non-zero packet count
- A traffic classification
- Security findings based on the real PCAP

8. If the run fails

On VM1:

```
tail -100 /home/kali/ESpect/testbed/sih-ipsec-analyzer_vm1/debug/logs/packet_capture.log
```

On VM1 and VM2:

```
sudo systemctl status espect-agent --no-pager -l  
sudo journalctl -u espect-agent --no-pager -n 80
```

Fix the first error in the log. Do not repeatedly click ****Start capture**** while a run is still marked **running**.

9. Stop for the day

Stop the Windows backend and frontend with **Ctrl+C** in their terminals. The VM services remain enabled and will start automatically next time the VMs boot.